

CSMMetal20250002 Designing Microwave-Proof Material

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMMetal20250002 Designing Microwave-Proof Material V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
ZrB2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Parameter	V1.0	V2.0	Reason	Primary Source
YInMn Blue spectral coating	V1.0: YInMn + 2 mol% Ta ₂ O ₅ ; SRI 115; NIR 85%	V2.0: CsPbBr ₃ QD co-deposit; SRI ~130; UV reflectance +31%	UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire	Solar Energy Mater. Sol. Cells 258 (2025) 112456
Structural ceramic option	V1.0: Monolithic ZrB ₂ -SiC only; diamond tooling required; brittle	V2.0: Ti ₃ AlC ₂ MAX Phase hybrid added; carbide-machinable; KNE quasi-ductile	35% machining cost reduction; eliminates brittle shatter at stress concentrations	Barsoum MAX Phases; Anasori & Gogotsi 2023

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
ZrB ₂ -SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH
YInMn Blue spectral coating	UV band protection gap; accelerated UV-induced substrate polymer degradation	MEDIUM
Structural ceramic option	Diamond tooling required; brittle fracture risk at fastener/joint stress concentrations	MEDIUM

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMMetal20250002 Designing Microwave-Proof Material V1.0 archived; V2.0 is the active specification as of June 2026.